

EDM OS — Commercial Engine

Build specification: payment applications, certification tracking and variation evidence.

WHY THIS EXISTS

EDM OS has no record of the commercial heartbeat of a subcontracting business: the monthly payment application, what the main contractor certified against it, the deductions and their reasons, and the payment clock. The Invoice model is an accounting object; this engine is the commercial one. It replaces the previously scoped generic pipeline work as build assignment three — the pipeline data models (Lead, Opportunity, Tender) already exist and need screens only.

MODEL 1 — PaymentApplication

One row per application per project per period. Cumulative by design: each application states gross value to date, and the system derives the period movement.

projectId / organisationId	FK	Multi-tenant, consistent with all models
applicationNo	Int, unique per project	Claim 1, 2, 3...
periodEnd	DateTime	Application period
grossAppliedToDate	Decimal(14,2)	Cumulative — the re-measure figure
submittedAt / certifiedAt / dueAt / paidAt	DateTime?	The full clock: terms policing from dueAt
grossCertifiedToDate	Decimal(14,2)?	What the QS certified, cumulative
retentionPct / retentionHeld	Decimal	Default from project; computed
paidAmount	Decimal(14,2)?	Supports partial payment
status	Enum	DRAFT / SUBMITTED / CERTIFIED / OVERDUE / PAID / DISPUTED
supportingDocs	Document[]	Measure quantities + Evidence QA packs attached

MODEL 2 — Deduction (child of PaymentApplication)

applicationId	FK	Which claim it was taken from
amount	Decimal(14,2)	Value knocked back
reasonCode	Enum	MEASURE / QUALITY / CONTRA / SETOFF / UNEXPLAINED
reasonText	String	The QS's stated reason, verbatim
status	Enum	ACCEPTED / CONTESTED / RECOVERED
resolvedInApplicationId	FK?	Where the money was won back

Every dirham refused gets a row, a reason and a fate. Uncontested deductions become precedent; this model makes silence impossible.

MODEL 3 — Variation (upgrade existing)

- Split value into submittedValue and agreedValue — the negotiation gap becomes visible and reportable.
- Add timeImpactDays (Int?) and costBreakdown (Json?) for labour/plant/material make-up.
- Add sourceEventId FK to SiteEvent — every variation traces to the Daybook instruction that caused it. This is the evidence chain: instruction logged on site becomes money certified, with nothing verbal in between.
- Status flow extended: DRAFT / SUBMITTED / AGREED / REJECTED / WITHDRAWN, with decidedAt retained.

MODEL 4 — Client scorecard (computed, not stored)

No manual fields. A nightly job (or on-demand query) derives per Company across all its projects:

- certificationRate — sum certified / sum applied, trailing 12 months.
- avgDaysToPay — mean of (paidAt - certifiedAt) against contractual terms.
- deductionProfile — deduction value by reasonCode, exposing how each MC behaves.

Surface on the Company page and in tender qualification: bid decisions informed by how the client actually pays, not how they promise to.

SCREENS — MINIMUM SET

- Claims ledger per project: one row per application, applied vs certified vs paid columns, ageing against dueAt, colour state on overdue.
- Application detail: cumulative build-up, attached Measure quantities and Evidence packs, deduction table with contest actions.
- Applied-vs-certified chart per project and per client — the Ajman chart, systemised.
- Variations register: submitted vs agreed totals, evidence link state, aged unpriced instructions flagged.
- Board rollup: group cash position — certified-unpaid, overdue, retention held, deductions in dispute.

ACCEPTANCE TESTS — DEFINITION OF DONE

- The full Ajman history (claims 1–5, the 80% certification pattern, the cumulative re-measure and contested deductions) can be entered and reproduces the known figures exactly.
- An application overdue against dueAt raises a Notification to the project owner without manual checking.
- A variation created from a SiteEvent carries its evidence link and appears in the next application's build-up.
- Client scorecard values recompute correctly when a payment lands; verified against hand-calculated figures.
- All new models carry organisationId, appear in AuditLog, and migrations run clean on the existing seed.

SEQUENCE AND GUARDRAILS

This is developer assignment three, after production hosting (one) and custody (two) per the engagement brief. Estimated effort: one to two retained months alongside those. No UI framework changes, no new services — extend the existing NestJS modules and Next.js app. Schema changes by Prisma migration only, reviewed before apply. The measure of success is simple: the day this ships, the Ajman claims story lives in EDM OS instead of a conversation.